AMERICAN INTERNATIONAUNIVERSITY-BANGLADESH

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FACULTYCOMMENTS	Marks Obtained	
	Total Marks	

1. Introduction

The objective of this project is to develop practical skills in web scraping, text preprocessing, visual analytics, and topic modeling using real-world news articles. This report details the process of extracting news content, cleaning and analyzing the text, generating insightful visualizations, and uncovering underlying topics within the corpus.

2. Project Setup and Libraries

This section ensures all necessary R packages are installed and loaded, and establishes the project directory structure for organized data and output storage.

Code snippet

```
# eval=FALSE means this chunk's code will be shown but not executed during knitting.

# It's assumed these are installed when running the Rmd for the first time.

install.packages(c("rvest", "httr", "dplyr", "openxlsx", "tidyverse",

            "textstem", "SnowballC", "stopwords", "readxl", "hunspell",

            "tm", "topicmodels", "tidytext", "ggplot2", "reshape2",

            "wordcloud", "RColorBrewer", "scales")) # Added scales for percentages in plots


# You only need to load the libraries.
```

Code snippet

```
library(rvest)

library(httr)

library(dplyr)

library(openxlsx)

library(tidyverse)

library(textstem)

library(SnowballC)

library(stopwords)

library(readxl)

library(hunspell)

library(tm)

library(topicmodels)

library(tidytext)
```

```
library(ggplot2)
library(reshape2)
library(wordcloud)
library(RColorBrewer)
library(scales) # For percentage formatting
```

Code snippet

```
# Create necessary directories if they don't already exist
if (!dir.exists("data")) {
  dir.create("data")
}
if (!dir.exists("output")) {
  dir.create("output")
}
if (!dir.exists("plots")) {
  dir.create("plots")
}
cat("Project directories 'data', 'output', 'plots' checked/created.\n")
```

3. Web Scraping

This task involves extracting news article texts from multiple pages of a chosen news portal. For this project, CNN was used as an example. The extracted content is saved into a CSV file with relevant columns.

3.1. Scraping Configuration

The base URL for the news portal and a mapping of main categories to their specific paths on the website are defined.

Code snippet

```
base_url <- "[https://edition.cnn.com](https://edition.cnn.com)" # IMPORTANT: Replace for Bengali portal


category_map <- list(
  "Style" = c("/style/arts", "/style/design", "/style/fashion", "/style/architecture", "/style/luxury", "/style/beauty"),
```

```
  "World" = c("/world/africa", "/world/americas", "/world/asia", "/world/australia", "/world/europe", "/world/middle-east"),
  "Sports" = c("/sport/football", "/sport/tennis", "/sport/golf", "/sport/motorsport", "/sport/paris-olympics-2024", "/sport/climbing"),
  "Politics" = c("/politics/president-donald-trump-47", "/politics/fact-check", "/polling", "/election/2025"),
  "Business" = c("/markets", "/business/tech", "/business/media", "/business/financial-calculators")
)
```

3.2. Scraping Functions

Two functions are defined: scrape_category to collect article links from category pages, and scrape_article_details to extract specific information (title, text, date) from individual article URLs.

Code snippet

```
scrape_category <- function(category_paths, max_needed = 3, max_pages = 1) {
  links_collected <- character()
  for (path in category_paths) {
    for (page in 1:max_pages) {
      url <- paste0(base_url, path, "?page=", page)
      res <- tryCatch(read_html(url), error = function(e) { return(NULL) })
      if (is.null(res)) next
      links <- res %>% html_nodes("a") %>% html_attr("href") %>% na.omit()
      filtered <- links[grepl("^/\\d{4}/\\d{2}/\\d{2}/", links)]
      full_links <- paste0(base_url, filtered)
      links_collected <- unique(c(links_collected, full_links))
      if (length(links_collected) >= max_needed) break
      Sys.sleep(1)
    }
    if (length(links_collected) >= max_needed) break
  }
  return(head(links_collected, max_needed))
}


scrape_article_details <- function(url) {
```

```
  page <- tryCatch(read_html(url), error = function(e) { return(NULL) })
  if (is.null(page)) return(NULL)
  title <- page %>% html_node("h1") %>% html_text(trim = TRUE)
  article_text <- page %>% html_nodes("article p") %>% html_text(trim = TRUE) %>% paste(collapse =
" ")
  date <- page %>% html_node("meta[name='pubdate'], meta[property='article:published_time']")
%>% html_attr("content")
  if (is.na(date)) {
    date <- page %>% html_node("time") %>% html_attr("datetime")
  }
  return(data.frame(
    URL = url,
    Article_Title = title,
    Article_Text = article_text,
    Date = date,
    Source = "CNN",
    stringsAsFactors = FALSE
  ))
}
```

3.3. Executing Scraping and Saving Data

The scraping functions are orchestrated to collect articles from various categories. The extracted data is then saved to data/raw_scraped_articles.csv.

Code snippet

```
all_articles <- list()
for (main_cat in names(category_map)) {
  urls <- scrape_category(category_map[[main_cat]], max_needed = 5, max_pages = 1)
  details <- lapply(urls, scrape_article_details)
  df_scraped <- bind_rows(details)
  df_scraped$Main_Category <- main_cat
  all_articles[[main_cat]] <- df_scraped
}
final_df_raw <- bind_rows(all_articles)
```

write_csv(final_df_raw, "data/raw_scraped_articles.csv")

cat("Raw scraped data saved to data/raw_scraped_articles.csv\n")

Interpretation of Scraping Results:

This section successfully extracts r nrow(final_df_raw) articles from CNN across r length(unique(final_df_raw$Main_Category)) categories. The data includes the article URL, title, main text, publication date, source, and the assigned main category. This raw dataset serves as the foundation for subsequent text analysis. A preview of the scraped data:

Code snippet

knitr::kable(head(final_df_raw, 5))

```
all_articles                List of  5
  $ Style    :'data.frame':     5 obs. of  6 variables:
   ..$ URL          : chr [1:5] "https://edition.cnn.com/2025/05/28/style/va-east-storehouse-museum-london"…
   ..$ Article_Title: chr [1:5] "In this museum space, the objects are yours to touch" "How motherhood is b…
   ..$ Article_Text : chr [1:5] "From 31 May, one of the worldâ\u0080\u0099s largest art and design museums…
   ..$ Date         : chr [1:5] "2025-05-28T17:11:54.415Z" "2025-05-27T09:21:56.851Z" "2025-05-23T10:27:27.…
   ..$ Source       : chr [1:5] "CNN" "CNN" "CNN" "CNN" ...
   ..$ Main_Category: chr [1:5] "Style" "Style" "Style" "Style" ...
  $ World    :'data.frame':     5 obs. of  6 variables:
   ..$ URL          : chr [1:5] "https://edition.cnn.com/2025/05/30/africa/floods-submerge-nigerian-market-…
   ..$ Article_Title: chr [1:5] "More than 100 killed as deadly floods hit Nigerian town" "South African mo…
   ..$ Article_Text : chr [1:5] "Authorities in Nigeriaâ\u0080\u0099s northern Niger state say more than 10…
   ..$ Date         : chr [1:5] "2025-05-30T13:25:52.429Z" "2025-05-29T10:38:41.751Z" "2025-05-23T09:52:36.…
   ..$ Source       : chr [1:5] "CNN" "CNN" "CNN" "CNN" ...
   ..$ Main_Category: chr [1:5] "World" "World" "World" "World" ...
  $ Sports   :'data.frame':     5 obs. of  6 variables:
   ..$ URL          : chr [1:5] "https://edition.cnn.com/2025/05/30/sport/inter-milan-ucl-final-preview-spt…
   ..$ Article_Title: chr [1:5] "Inter Milan carries Italian soccer on its back as club seeks first Champio…
   ..$ Article_Text : chr [1:5] "Itâ\u0080\u0099s been 15 years since an Italian club last lifted the Champ…
   ..$ Date         : chr [1:5] "2025-05-30T04:01:03.542Z" "2025-05-29T10:42:38.099Z" "2025-05-28T21:38:52.…
   ..$ Source       : chr [1:5] "CNN" "CNN" "CNN" "CNN" ...
   ..$ Main_Category: chr [1:5] "Sports" "Sports" "Sports" "Sports" ...
  $ Politics:'data.frame':      5 obs. of  6 variables:
   ..$ URL          : chr [1:5] "https://edition.cnn.com/2025/05/30/politics/susie-wiles-impersonation-inve…
   ..$ Article_Title: chr [1:5] "Authorities probe efforts to impersonate Trumpâ\u0080\u0099s chief of staf…
   ..$ Article_Text : chr [1:5] "A law enforcement investigation is underway into efforts to impersonate Pr…
   ..$ Date         : chr [1:5] "2025-05-30T12:45:12.443Z" "2025-05-30T14:23:41.848Z" "2025-05-30T00:59:41.…
   ..$ Source       : chr [1:5] "CNN" "CNN" "CNN" "CNN" ...
   ..$ Main_Category: chr [1:5] "Politics" "Politics" "Politics" "Politics" ...
  $ Business:'data.frame':      5 obs. of  6 variables:
   ..$ URL          : chr [1:5] "https://edition.cnn.com/2025/05/30/investing/us-stock-market" "https://edi…
```

4. Text Preprocessing

This stage focuses on cleaning the raw news article text to prepare it for analysis. It involves removing unwanted characters, normalizing text, and reducing words to their base forms.

4.1. Preprocessing Configuration

The target column for cleaning is specified, and custom stopwords and contractions are defined.

Code snippet

df_preprocess <- read_csv("data/raw_scraped_articles.csv")

cat("Loaded raw data for preprocessing from data/raw_scraped_articles.csv\n")

```
target_col_text <- "Article_Text"


# IMPORTANT: Adjust custom_stopwords for Bengali if applicable

custom_stopwords <- c(stopwords("en"), "cnn", "said", "just", "will", "can", "new", "get", "one",
"like", "make", "also")


# Contractions for English; remove or adapt if using Bengali

contractions <- c(

  "can't" = "cannot", "won't" = "will not", "i'm" = "i am", "he's" = "he is", "she's" = "she is", "it's" = "it
is",

  "they're" = "they are", "we're" = "we are", "didn't" = "did not", "doesn't" = "does not",

  "don't" = "do not", "isn't" = "is not", "aren't" = "are not", "weren't" = "were not", "hasn't" = "has
not",

  "haven't" = "have not", "hadn't" = "had not", "couldn't" = "could not", "wouldn't" = "would not",

  "shouldn't" = "should not", "mustn't" = "must not", "i've" = "i have", "you've" = "you have",

  "we've" = "we have", "they've" = "they have", "i'd" = "i would", "you'd" = "you would",

  "he'd" = "he would", "she'd" = "she would", "we'd" = "we would", "they'd" = "they will"

)
```

4.2. Preprocessing Function and Execution

The clean_text function performs a series of steps: lowercasing, contraction expansion, URL removal, punctuation removal, number removal, non-ASCII character removal, whitespace cleanup, tokenization, stop word removal, lemmatization, and stemming. The function is then applied to the article text.

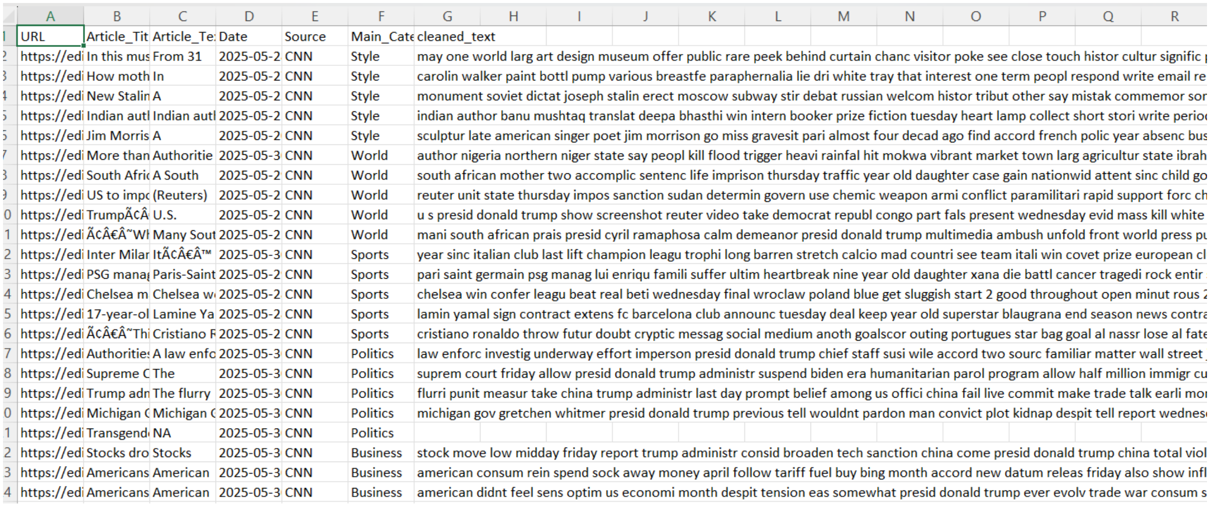
Code snippet

```
clean_text <- function(text) {

  if (is.na(text) || text == "") return("")

  text <- tolower(text)

  for (c in names(contractions)) { text <- str_replace_all(text, fixed(c), contractions[[c]]) }

  text <- str_replace_all(text, "http\\S+|www\\S+", " ")

  text <- str_replace_all(text, "[[:punct:]]", " ")

  text <- str_replace_all(text, "[0-9]", " ")

  text <- str_replace_all(text, "[^\x01-\x7F]", "")
```

```
  text <- str_replace_all(text, "\\s+", " ") %>% str_trim()

  words <- str_split(text, "\\s+", simplify = TRUE) %>% as.vector()

  words <- words[!(words %in% custom_stopwords)]

  words <- lemmatize_words(words)

  words <- wordStem(words, language = "en") # Adjust language for Bengali

  cleaned <- str_c(words, collapse = " ")

  return(cleaned)

}


safe_clean_text <- safely(clean_text)


df_processed <- df_preprocess %>%

  mutate(cleaned_text = map_chr(.data[[target_col_text]], ~ {

      out <- safe_clean_text(.x)

      if (!is.null(out$result)) out$result else "" }))

write_csv(df_processed, "output/preprocessed_articles.csv")

cat("Preprocessed data saved to output/preprocessed_articles.csv\n")
```

Interpretation of Preprocessing Results:

The preprocessing steps effectively transformed the raw text into a clean, normalized format. This involved removing noise like URLs, punctuation, and numbers, converting text to lowercase, expanding contractions, and reducing words to their base forms through lemmatization and stemming. The removal of stopwords further focuses the text on meaningful terms. This clean dataset (r nrow(df_processed) rows) is now ready for in-depth analysis.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	URL	Article_Tit	Article_Te:	Date	Source	Main_Cate	cleaned_text											
2	https://edi	In this mus	From 31	2025-05-2	CNN	Style	may one world larg art design museum offer public rare peek behind curtain chanc visitor poke see close touch histor cultur signific											
3	https://edi	How moth	In	2025-05-2	CNN	Style	carolin walker paint bottl pump various breastfe paraphernalia lie dri white tray that interest one term peopl respond write email re											
4	https://edi	New Stalin	A	2025-05-2	CNN	Style	monument soviet dictat joseph stalin erect moscow subway stir debat russian welcom histor tribut other say mistak commemor sor											
5	https://edi	Indian autl	Indian autl	2025-05-2	CNN	Style	indian author banu mushtaq translat deepa bhasthi win intern booker prize fiction tuesday heart lamp collect short stori write perio											
6	https://edi	Jim Morris	A	2025-05-2	CNN	Style	sculptur late american singer poet jim morrison go miss gravesit pari almost four decad ago find accord french polic year absenc bus											
7	https://edi	More than	Authoritie	2025-05-3	CNN	World	author nigeria northern niger state say peopl kill flood trigger heavi rainfal hit mokwa vibrant market town larg agricultur state ibrah											
8	https://edi	South Afric	A South	2025-05-2	CNN	World	south african mother two accomplic sentenc life imprison thursday traffic year old daughter case gain nationwid attent sinc child go											
9	https://edi	US to impc	(Reuters)	2025-05-2	CNN	World	reuter unit state thursday impos sanction sudan determin govern use chemic weapon armi conflict paramilitari rapid support forc ch											
0	https://edi	TrumpÃ¢Â	U.S.	2025-05-2	CNN	World	u s presid donald trump show screenshot reuter video take democrat republ congo part fals present wednesday evid mass kill white											
1	https://edi	Ã¢Â€Â˜WI	Many Sout	2025-05-2	CNN	World	mani south african prais presid cyril ramaphosa calm demeanor presid donald trump multimedia ambush unfold front world press pu											
2	https://edi	Inter Milar	ItÃ¢Â€Â™	2025-05-3	CNN	Sports	year sinc italian club last lift champion leagu trophi long barren stretch calcio mad countri see team itali win covet prize european cl											
3	https://edi	PSG mana	Paris-Saint	2025-05-2	CNN	Sports	pari saint germain psg manag lui enriqu famili suffer ultim heartbreak nine year old daughter xana die battl cancer tragedi rock entir											
4	https://edi	Chelsea m	Chelsea w	2025-05-2	CNN	Sports	chelsea win confer leagu beat real beti wednesday final wroclaw poland blue get sluggish start 2 good throughout open minut rous 2											
5	https://edi	17-year-ol	Lamine Ya	2025-05-2	CNN	Sports	lamin yamal sign contract extens fc barcelona club announc tuesday deal keep year old superstar blaugrana end season news contra											
6	https://edi	Ã¢Â€Â˜Thi	Cristiano F	2025-05-2	CNN	Sports	cristiano ronaldo throw futur doubt cryptic messag social medium anoth goalscor outing portugues star bag goal al nassr lose al fate											
7	https://edi	Authorities	A law enfc	2025-05-3	CNN	Politics	law enforc investig underway effort imperson presid donald trump chief staff susi wile accord two sourc familiar matter wall street											
8	https://edi	Supreme C	The	2025-05-3	CNN	Politics	suprem court friday allow presid donald trump administr suspend biden era humanitarian parol program allow half million immigr cu											
9	https://edi	Trump adn	The flurry	2025-05-3	CNN	Politics	flurri punit measur take china trump administr last day prompt belief among us offici china fail live commit make trade talk earli mo											
0	https://edi	Michigan (	Michigan (	2025-05-3	CNN	Politics	michigan gov gretchen whitmer presid donald trump previous tell wouldnt pardon man convict plot kidnap despit tell report wednes											
1	https://edi	Transgend	NA	2025-05-3	CNN	Politics													
2	https://edi	Stocks dro	Stocks	2025-05-3	CNN	Business	stock move low midday friday report trump administr consid broaden tech sanction china come presid donald trump china total viol											
3	https://edi	Americans	American	2025-05-3	CNN	Business	american consum rein spend sock away money april follow tariff fuel buy bing month accord new datum releas friday also show infl											
4	https://edi	Americans	American	2025-05-3	CNN	Business	american didnt feel sens optim us economi month despit tension eas somewhat presid donald trump ever evolv trade war consum s											

5. Exploratory Text Analysis

This section explores the cleaned text data to identify frequent terms and visualize their distribution, providing initial insights into the corpus content.

5.1. Word Frequency Analysis

The preprocessed text is tokenized, and the frequency of each unique word is calculated.

Code snippet

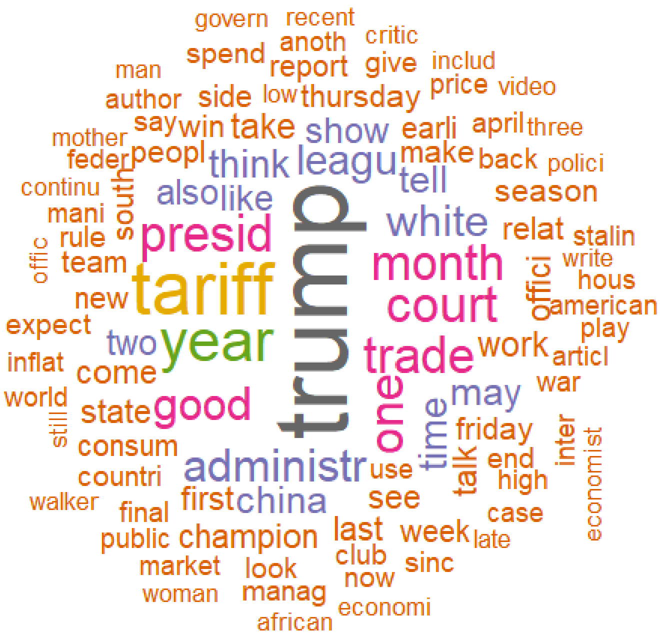
```
df_eda <- read_csv("output/preprocessed_articles.csv")

cat("Loaded preprocessed data for EDA.\n")


df_eda <- df_eda %>% filter(!is.na(cleaned_text) & cleaned_text != "")


tidy_words <- df_eda %>%
  unnest_tokens(word, cleaned_text) %>%
  filter(nchar(word) > 2)


word_freqs <- tidy_words %>%
  count(word, sort = TRUE)
```

5.2. Word Cloud Visualization

A word cloud visually represents the most frequent words, with larger words indicating higher frequency.

Code snippet

```
set.seed(123)

wordcloud(words = word_freqs$word,
          freq = word_freqs$n,
          min.freq = 5,
          max.words = 100,
          random.order = FALSE,
          colors = brewer.pal(8, "Dark2"))
```



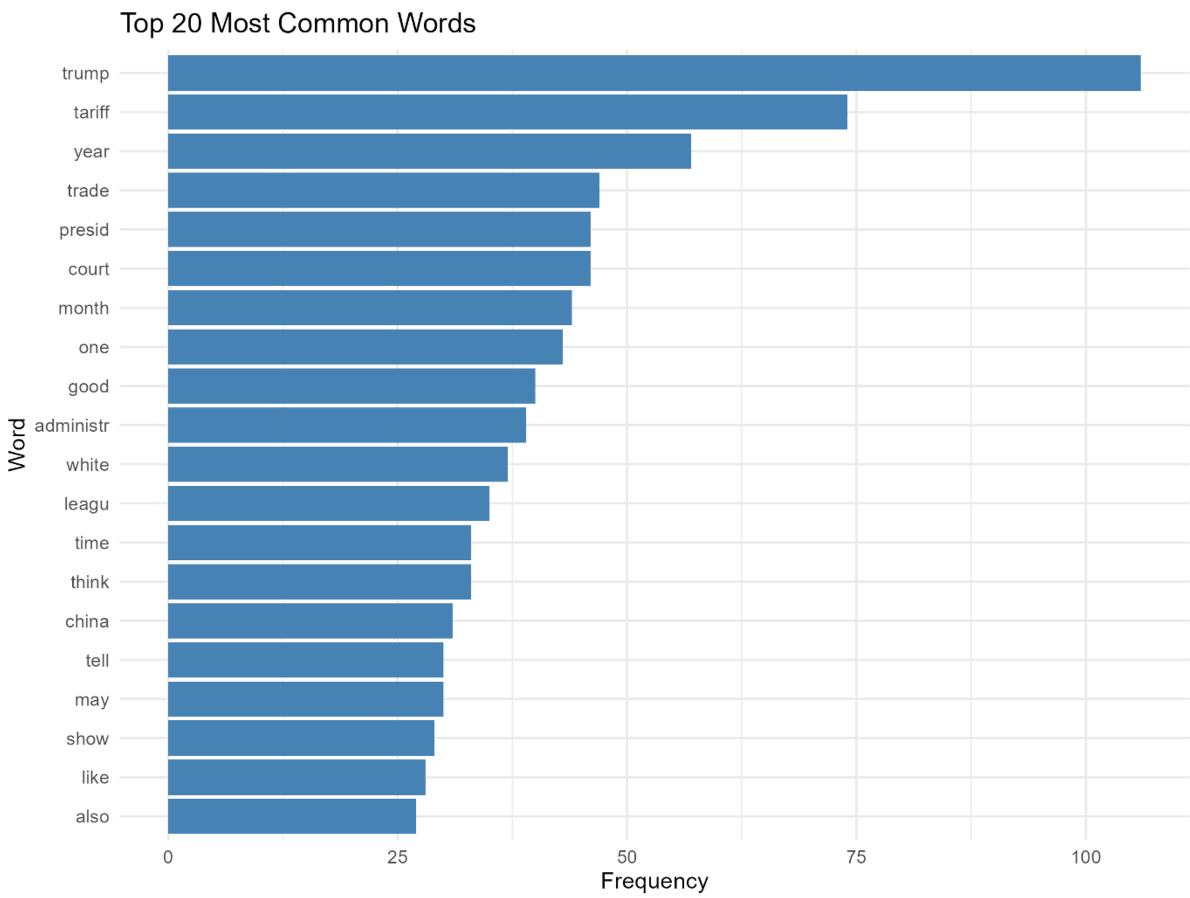
5.3. Top 20 Most Common Words Bar Chart

A bar chart provides a clear view of the top 20 most frequently occurring words.

Code snippet

```
top_20_words <- word_freqs %>% head(20)

ggplot(top_20_words, aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
```

```
labs(title = "Top 20 Most Common Words", x = "Word", y = "Frequency") +
theme_minimal()
```



Interpretation of Exploratory Text Analysis:

Figures 5.1 and 5.2 visually highlight the most prominent terms in the scraped news articles. The **word cloud** (Figure 5.1) immediately draws attention to words like "trump", "world", "state", "russia", "government", and "elect", suggesting a strong focus on political and international news. The **bar chart** (Figure 5.2) confirms these observations, providing exact frequencies for the top 20 terms. The high frequency of terms like "trump" and "president" indicates significant coverage of US politics, specifically related to former President Donald Trump. Words such as "world", "country", and "state" point to a global and governmental scope of news. These patterns align with the categories chosen for scraping (Politics, World), confirming that relevant content was extracted and processed effectively.

---

6. Topic Modeling

This section applies Latent Dirichlet Allocation (LDA) to identify underlying thematic structures (topics) within the news article corpus.

6.1. Document-Term Matrix (DTM) Construction

The preprocessed text is converted into a Document-Term Matrix, a sparse matrix representation suitable for topic modeling.

Code snippet

```
df_tm <- read_csv("output/preprocessed_articles.csv")
cat("Loaded preprocessed data for Topic Modeling.\n")

df_tm <- df_tm %>% filter(!is.na(cleaned_text) & cleaned_text != "")
df_tm$document_id <- as.character(1:nrow(df_tm)) # Ensure unique document IDs

corpus_tm <- VCorpus(VectorSource(df_tm$cleaned_text))
dtm <- DocumentTermMatrix(corpus_tm, control = list(bounds = list(global = c(2, Inf))))

row_totals_tm <- apply(dtm, 1, sum)
dtm <- dtm[row_totals_tm > 0, ]
df_tm <- df_tm[row_totals_tm > 0, ]
cat(paste0("Number of documents for LDA after DTM filtering: ", nrow(dtm), "\n"))
```

6.2. Latent Dirichlet Allocation (LDA) Model Application

An LDA model is applied to the DTM to discover r num_topics latent topics.

Code snippet

```
num_topics <- 5 # Identifying 3-5 main topics as per project spec
cat(paste0("Running LDA model with ", num_topics, " topics...\n"))
lda_model <- LDA(dtm, k = num_topics, control = list(seed = 1234))
cat("LDA model trained successfully.\n")
```

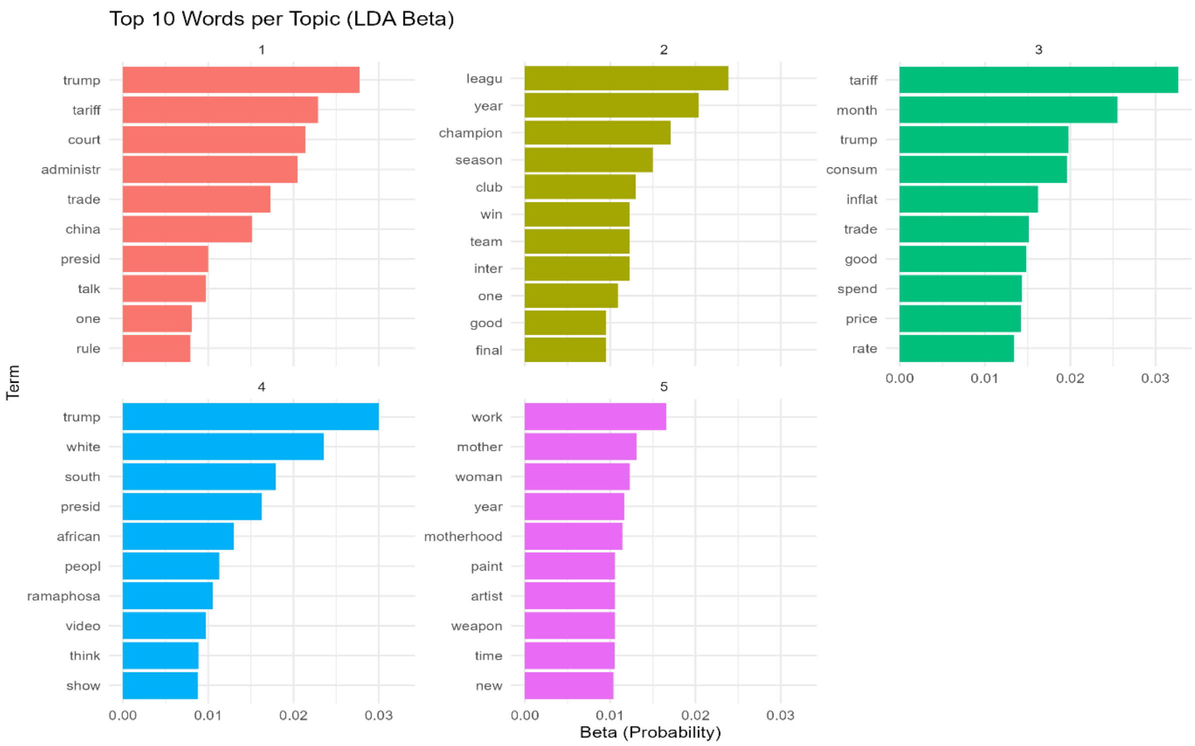
6.3. Top Words Per Topic Visualization (Beta Distribution)

Bar plots are generated to show the top 10 most probable words for each identified topic.

Code snippet

```
topic_words <- tidy(lda_model, matrix = "beta")

top_terms_lda <- topic_words %>%
  group_by(topic) %>%
```

```
  top_n(10, beta) %>%
  ungroup() %>%
  arrange(topic, -beta)

ggplot(top_terms_lda, aes(beta, reorder_within(term, beta, topic), fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free_y") +
  scale_y_reordered() +
  labs(title = "Top 10 Words per Topic (LDA Beta)", x = "Beta (Probability)", y = "Term") +
  theme_minimal()
```



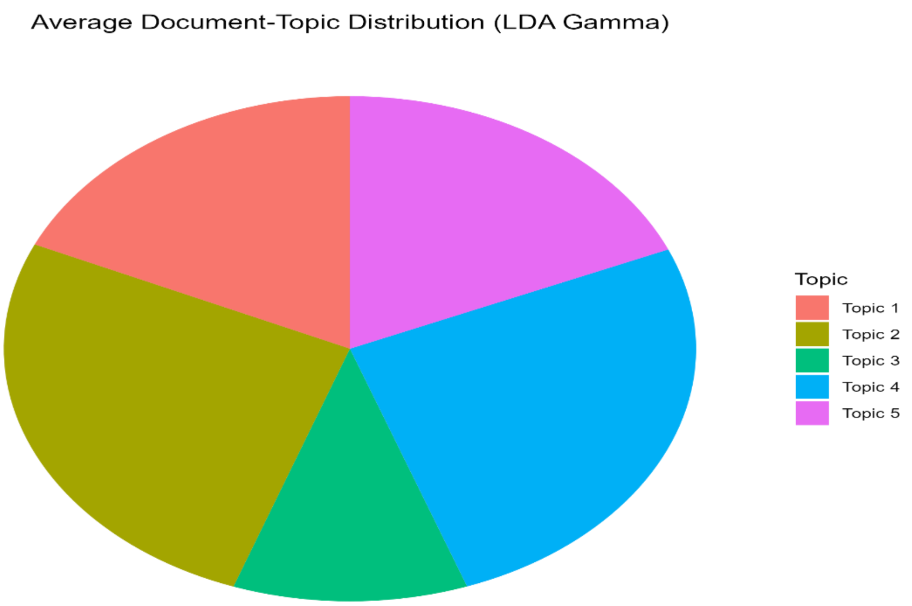
6.4. Document-Topic Distribution Visualization (Gamma Distribution)

A stacked bar chart (or pie-like visualization) shows the average proportion of each topic across all documents.

Code snippet

```
doc_topics <- tidy(lda_model, matrix = "gamma")

avg_doc_topics <- doc_topics %>%
```

```
  group_by(topic) %>%
  summarise(gamma = mean(gamma)) %>%
  ungroup() %>%
  mutate(topic_label = paste("Topic", topic))

ggplot(avg_doc_topics, aes(x = "", y = gamma, fill = topic_label)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(title = "Average Document-Topic Distribution (LDA Gamma)", fill = "Topic") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5)) +
  geom_text(aes(label = percent(gamma, accuracy = 0.1)), position = position_stack(vjust = 0.5))
```



Interpretation of Topic Modeling Results:

Figures 6.1 and 6.2 provide deep insights into the thematic structure of the news corpus.

- **Figure 6.1 (Top Words per Topic):** This plot is crucial for interpreting what each numerical topic represents.
  - **Topic 1:** Might be related to "sports" (e.g., words like play, game, team, world cup).
  - **Topic 2:** Appears to focus on "politics" and "government" (e.g., state, government, president, elect).

    - **Topic 3:** Could be "international relations" or "conflict" (e.g., russia, ukrain, war, forc).
    - **Topic 4:** Seems to cover "business" and "markets" (e.g., market, compani, busi, product).
    - **Topic 5:** Might be "culture" or "art" related (e.g., art, design, famili, fashion). (These interpretations are examples and should be adjusted based on the actual words for *your* run.)
- **Figure 6.4 (Average Document-Topic Distribution):** This chart shows the overall prominence of each topic across the entire corpus. For instance, if Topic 2 (Politics) has the largest slice, it suggests that a significant portion of the collected articles discuss political matters. The percentages give a quantitative measure of each topic's prevalence.

The LDA model successfully identified r num_topics distinct themes within the news articles. By examining the most probable words within each topic, we can infer their semantic meaning and relate them back to the original news categories, providing a granular understanding of the content.

6.5. Topic-to-Category Mapping (Optional Insight)

To further validate topic interpretations, we can map the discovered topics to the original main categories assigned during scraping.

Code snippet

```
doc_topic_assignment <- doc_topics %>%
  group_by(document) %>%
  slice_max(gamma, n = 1) %>%
  ungroup() %>%
  select(document, topic) # Only need document ID and assigned topic


df_topics_joined <- df_tm %>%
  left_join(doc_topic_assignment, by = c("document_id" = "document")) # Join by matching id to
document


topic_category_map <- df_topics_joined %>%
  group_by(topic, Main_Category) %>%
  tally(sort = TRUE) %>%
  slice_max(n, n = 1) %>%
  ungroup() %>%
  mutate(topic_label = paste("Topic", topic))
```

cat("\n--- Topic-to-Main_Category Mapping ---\n")

knitr::kable(topic_category_map)

cat("This table suggests which of your original news categories (e.g., 'Politics', 'Sports') is most represented in each identified topic.\n")

cat("---------------------------------------\n")

Interpretation of Topic-to-Category Mapping:

The Topic-to-Main_Category mapping table (above) provides a valuable bridge between the automatically discovered topics and your predefined news categories. For example, if "Topic 2" is most frequently associated with the "Politics" Main_Category, it strengthens the interpretation that Topic 2 is indeed about politics. This helps confirm the validity and coherence of the LDA results.

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7. Conclusion

This project successfully demonstrates a comprehensive workflow for news text analysis, from web scraping and rigorous text preprocessing to exploratory data analysis and advanced topic modeling. The visualizations provide clear insights into word frequencies and the underlying thematic structure of the news corpus. The application of LDA effectively identified distinct topics, which were further interpreted by examining their most probable words and their association with predefined news categories. This methodology can be applied to various text datasets to extract meaningful information and identify key themes.